

INTRUSION DETECTION SYSTEM DEPLOYMENT AND PACKET ANALYSIS USING SURICATA

INTRODUCTION

With the increasing sophistication of cyber threats, organizations require effective mechanisms to monitor and detect malicious network activities. Intrusion Detection Systems (IDS) play a critical role in identifying suspicious traffic and generating alerts for analysis. This project demonstrates the deployment of Suricata IDS in a controlled lab environment, simulation of network attacks, and analysis of captured traffic using Wireshark. The objective is to understand how IDS detects malicious behavior and how packet-level analysis supports security investigations.

AIM AND OBJECTIVES

AIM

To deploy and configure an Intrusion Detection System and analyze malicious network activities.

Objectives

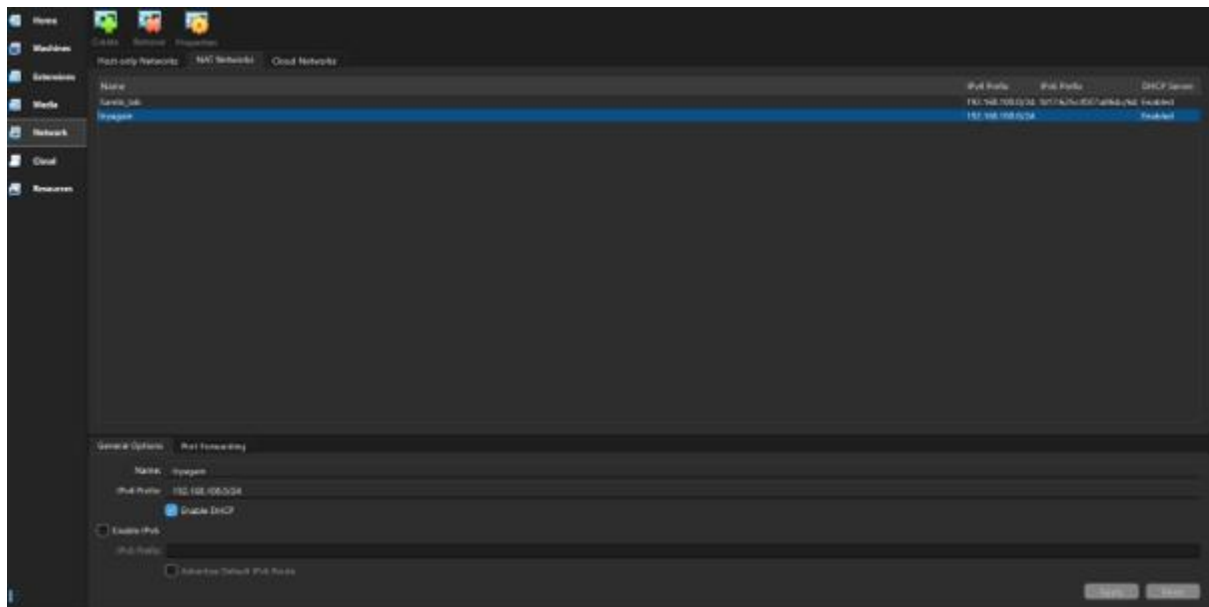
- Configure Suricata IDS in a Linux environment
 - Simulate reconnaissance and application-layer attacks
 - Detect malicious traffic using IDS rules
 - Capture and analyze packets using Wireshark
 - Correlate IDS alerts with real network traffic
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LAB SETUP AND ENVIRONMENT

The lab environment consists of:

- Kali Linux (Attacker + IDS + Analysis system)
- Metasploitable2 (Vulnerable target system)
- Virtual network configured using NAT (192.168.100.0/24 subnet)

This setup enables controlled communication between attacker and target systems.



```
(sambi@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d3:a0:15 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.4/24 brd 192.168.100.255 scope global dynamic eth0
        valid_lft 498sec preferred_lft 498sec
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:03:4f brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 570sec preferred_lft 570sec
    inet6 fe80::a00:27ff:febc:34f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:32:9b:a6:b3 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

✦ Figure 1: Network configuration / IP setup

SURICATA IDS DEPLOYMENT

Suricata was installed and configured on Kali Linux. The configuration file was modified to define the local network range:

HOME_NET: "[192.168.100.0/24]"

Suricata was executed in IDS mode using the command:

`sudo suricata -c /etc/suricata/suricata.yaml -i eth0`

This enabled real-time monitoring of network traffic on the selected interface.

```
(sambi@kali)-[~]
$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0
i: suricata: This is Suricata version 8.0.2 RELEASE running in SYSTEM mode
i: mpm-hs: Rule group caching - loaded: 114 newly cached: 0 total cacheable: 114
i: threads: Threads created -> W: 2 FM: 1 FR: 1 Engine started.
```

✦ Figure 2: Suricata running in terminal

ATTACK SIMULATION AND DETECTION

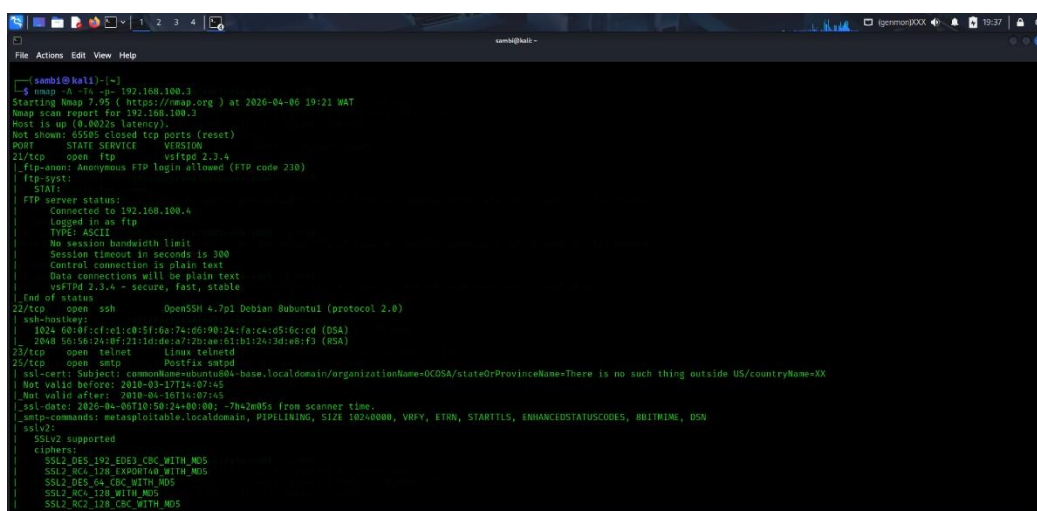
NMAP RECONNAISSANCE SCAN

An aggressive full port scan was performed using:

```
nmap -A -T4 -p- 192.168.100.3
```

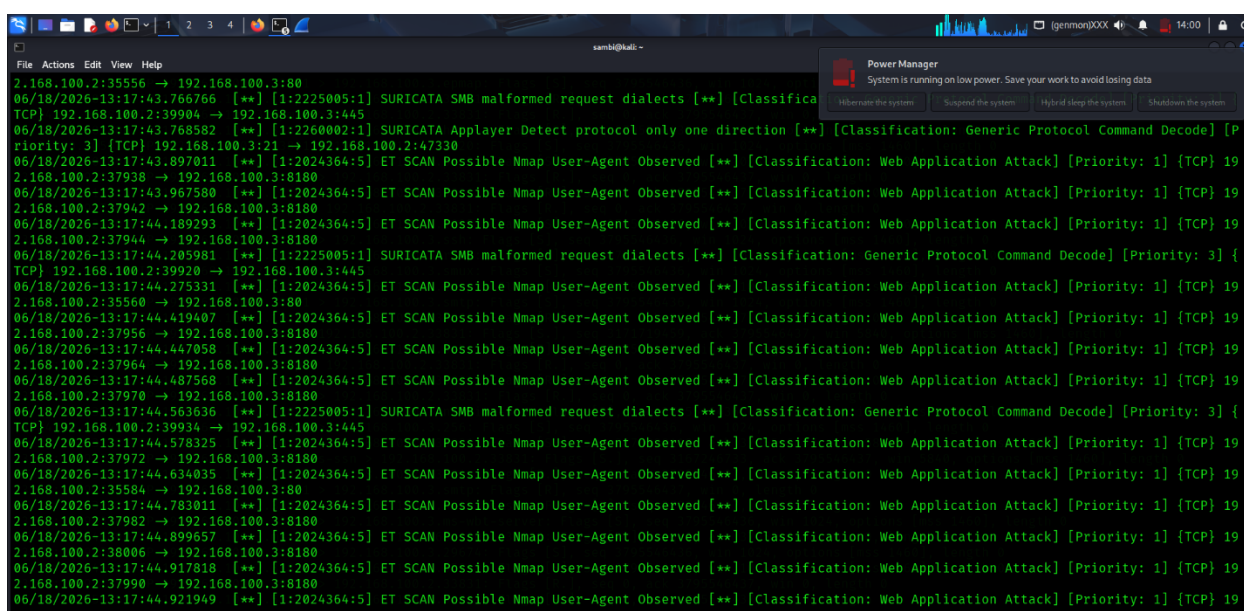
This scan attempts to identify open ports and services on the target system.

Suricata detected this activity and generated alerts indicating suspicious scanning behavior.



```
[sam@kali:~]$ nmap -A -T4 -p- 192.168.100.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-06 19:21 WAT
Nmap scan report for 192.168.100.3
Host is up (0.0022s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_ftp-anon Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.100.4
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_end of status
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
ssh-hostkey:
|_1024 60:0f:cf:ef:1c:5f:6a:7a:66:90:24:f0:c0:d5:0c:cd (DSA)
|_1024 56:56:24:0f:21:1d:de:a7:2b:0e:61:b1:2a:3d:0b:f3 (RSA)
23/tcp    open  telnet   Linux telnetd
23/tcp    open  netcat   Postfix netcat
|_ssl-cert: Subject: commonName=subunit004-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
Not valid before: 2018-03-17T14:07:45
Not valid after: 2018-04-10T14:07:45
ssl-date: 2026-04-06T19:24:00.000Z-7ba2m855 from scanner time.
|_nmap-command: metasploitable.localdomain, PIPELINING, SIZE 1024x8080, VERIFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, BBIMTIME, DSN
sslv2:
|_SSLv2 supported
ciphers:
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_SSL2_RSA_128_EXPORT1024_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_SSL2_RSA_128_WITH_MD5
|_SSL2_RSA_128_EXPORT1024_WITH_MD5
```

✦ Figure 3: Nmap scan execution



```
2.168.100.2:35556 → 192.168.100.3:80
06/18/2026-13:17:43.766766 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.100.2:35556 → 192.168.100.3:80
06/18/2026-13:17:43.768582 [**] [1:2260002:1] SURICATA Applayer Detect protocol only one direction [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.100.2:35556 → 192.168.100.3:80
06/18/2026-13:17:43.897011 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37938 → 192.168.100.3:8180
06/18/2026-13:17:43.967580 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37942 → 192.168.100.3:8180
06/18/2026-13:17:44.189293 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37944 → 192.168.100.3:8180
06/18/2026-13:17:44.205981 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.100.2:39920 → 192.168.100.3:445
06/18/2026-13:17:44.275331 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:35560 → 192.168.100.3:80
06/18/2026-13:17:44.419407 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37956 → 192.168.100.3:8180
06/18/2026-13:17:44.447058 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37964 → 192.168.100.3:8180
06/18/2026-13:17:44.487568 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37970 → 192.168.100.3:8180
06/18/2026-13:17:44.563636 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.100.2:39934 → 192.168.100.3:445
06/18/2026-13:17:44.578325 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37972 → 192.168.100.3:8180
06/18/2026-13:17:44.634035 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:35584 → 192.168.100.3:80
06/18/2026-13:17:44.783011 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37982 → 192.168.100.3:8180
06/18/2026-13:17:44.899657 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:38006 → 192.168.100.3:8180
06/18/2026-13:17:44.917818 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37990 → 192.168.100.3:8180
06/18/2026-13:17:44.921949 [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.100.2:37990 → 192.168.100.3:8180
```

✦ Figure 4: Suricata alert for Nmap scan

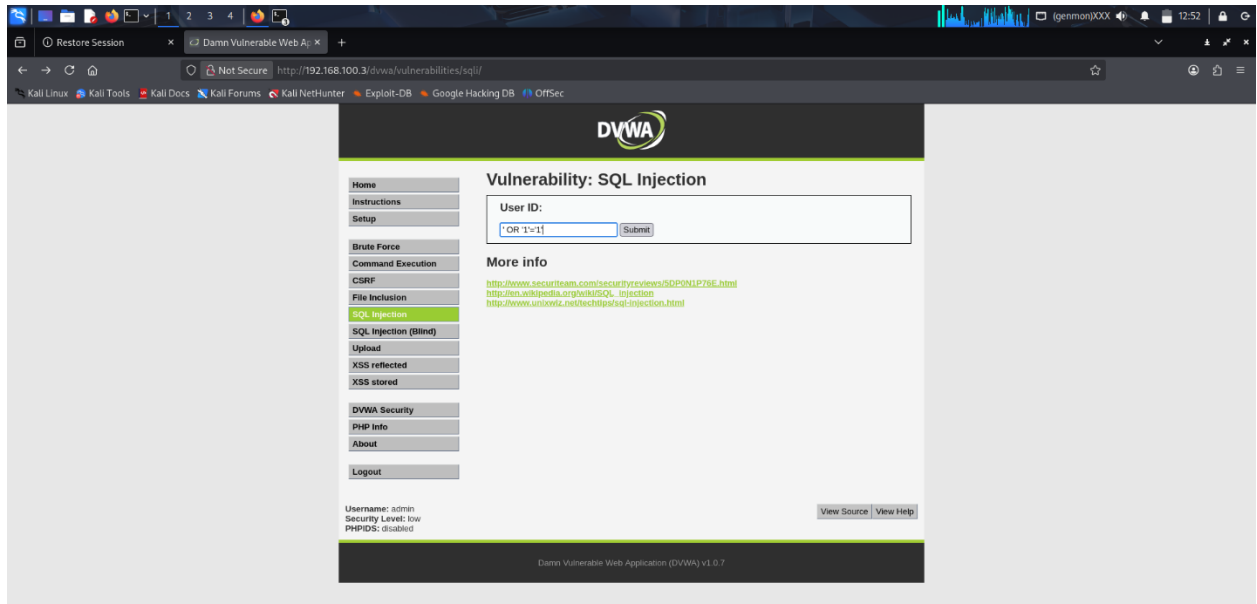
SQL INJECTION SIMULATION

A SQL Injection attack was performed on DVWA using the payload:

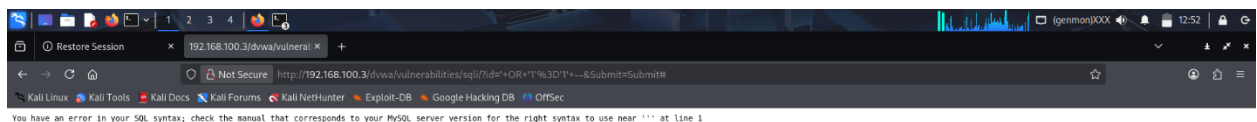
' OR '1'='1' --

This payload manipulates SQL queries to bypass authentication and retrieve database records, demonstrating improper input validation.

Although initial attempts generated errors, this confirmed that user input was directly processed by the backend database.



✦ *Figure 5: DVWA SQL injection input*



✦ *Figure 6: Error or result page*

PACKET CAPTURE AND ANALYSIS (WIRESHARK)

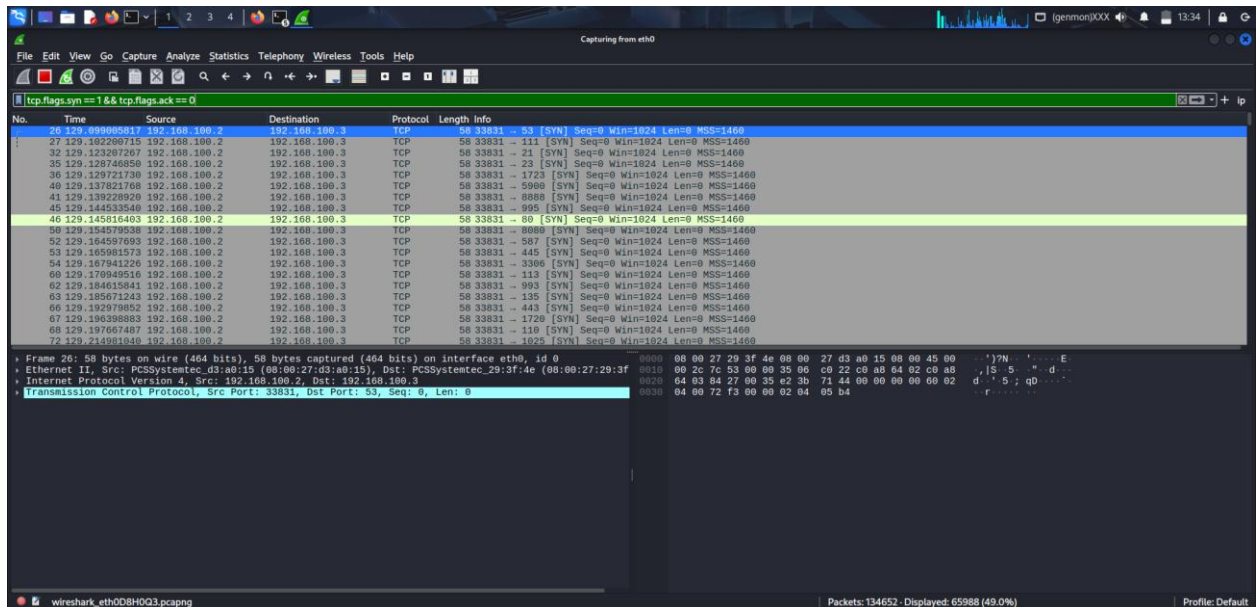
Wireshark was used to capture and analyze network traffic during attack simulations.

Nmap Traffic Analysis

Filter used:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

Multiple SYN packets were observed, confirming TCP port scanning activity.



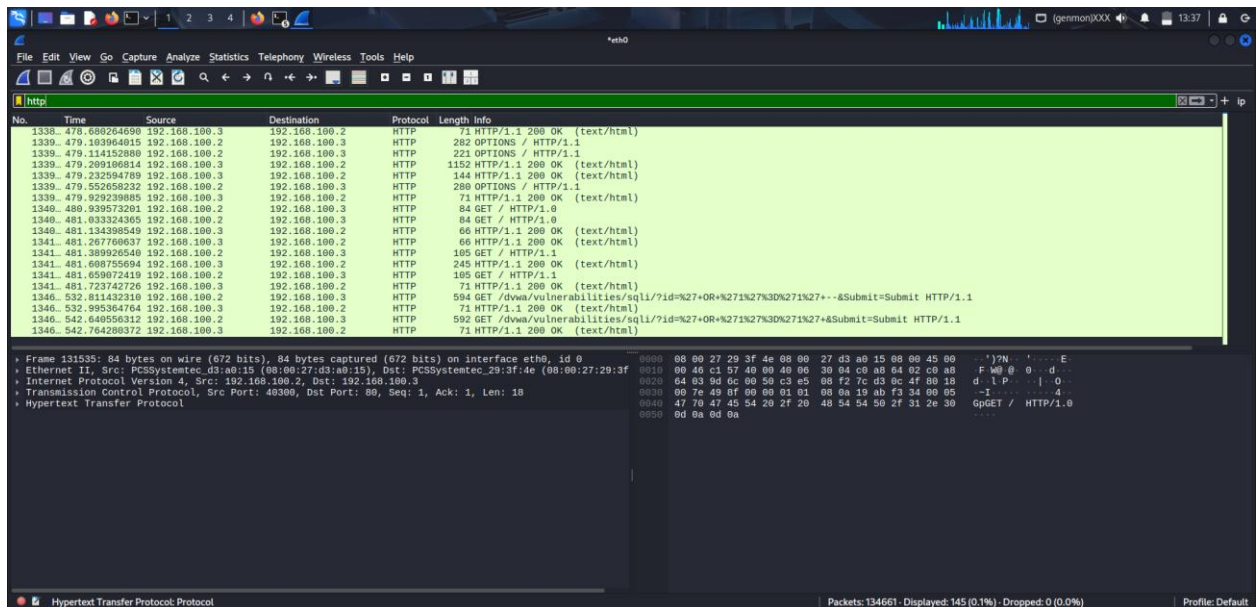
✦ Figure 7: SYN packets in Wireshark

SQL INJECTION TRAFFIC ANALYSIS

Filter used:

```
http
```

HTTP requests containing SQL injection payloads were captured, confirming malicious input transmission.



✦ Figure 8: HTTP packet showing SQL payload

CORRELATION OF IDS ALERTS AND TRAFFIC

A key aspect of this project is correlating IDS alerts with actual network traffic.

- Suricata generated alerts for scanning and web-based attacks
- Wireshark confirmed corresponding packet-level activity
- Source and destination IP addresses matched across both tools
- Payload analysis validated malicious behavior

This demonstrates the effectiveness of IDS in real-time threat detection.

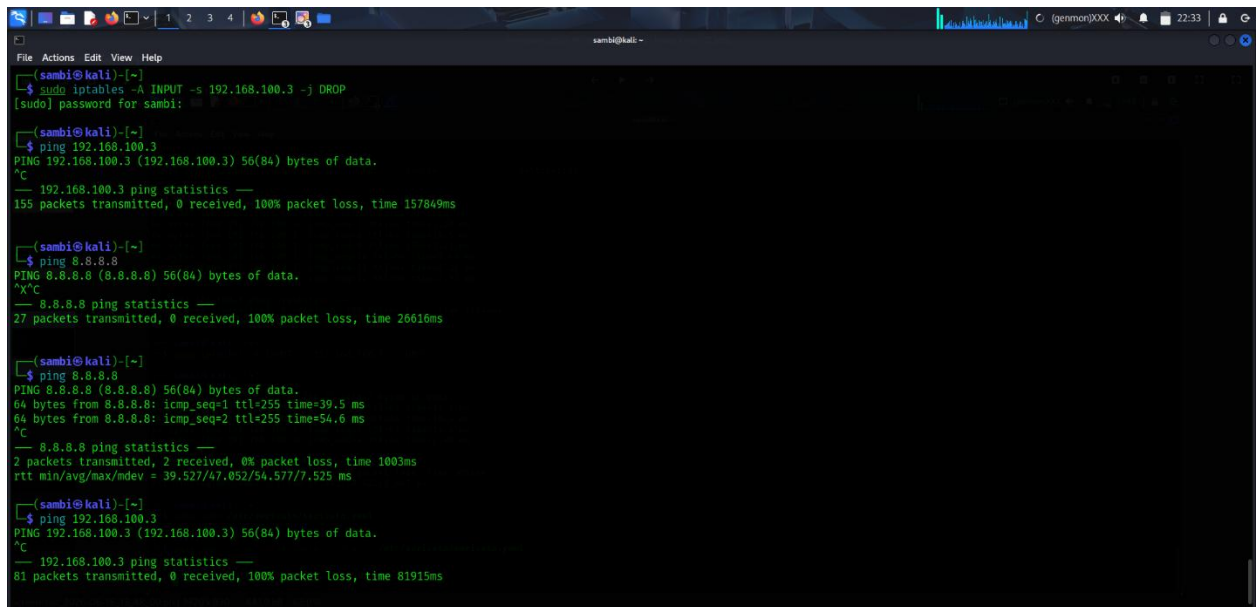
RESPONSE MECHANISM

To demonstrate basic intrusion response, malicious traffic sources can be blocked using firewall rules.

Example:

```
sudo iptables -A INPUT -s 192.168.100.3 -j DROP
```

This prevents further communication from the identified attacker IP.

A terminal window on a Kali Linux system. The user is at the prompt 'sambi@kali:~'. They run 'sudo iptables -A INPUT -s 192.168.100.3 -j DROP'. Then they run 'ping 192.168.100.3', which shows 100% packet loss. Next, they run 'ping 8.8.8.8', which shows 100% packet loss. Finally, they run 'ping 8.8.8.8' again, which shows successful connectivity with 0% packet loss. The terminal output is as follows:

```
(sambi@kali)~$ sudo iptables -A INPUT -s 192.168.100.3 -j DROP
[sudo] password for sambi:
(sambi@kali)~$ ping 192.168.100.3
PING 192.168.100.3 (192.168.100.3) 56(84) bytes of data.
^C
--- 192.168.100.3 ping statistics ---
155 packets transmitted, 0 received, 100% packet loss, time 157849ms

(sambi@kali)~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^X^C
--- 8.8.8.8 ping statistics ---
27 packets transmitted, 0 received, 100% packet loss, time 26616ms

(sambi@kali)~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=39.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=54.6 ms
^C
--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 39.527/47.052/54.577/7.525 ms

(sambi@kali)~$ ping 192.168.100.3
PING 192.168.100.3 (192.168.100.3) 56(84) bytes of data.
^C
--- 192.168.100.3 ping statistics ---
81 packets transmitted, 0 received, 100% packet loss, time 81915ms
```

✦ *Figure 9: iptables rule execution*

FINDINGS

- Suricata effectively detected reconnaissance activities such as Nmap scans
- HTTP-based attacks such as SQL injection were successfully identified
- Wireshark provided detailed packet-level insight into attack behavior
- Improper input validation exposes web applications to serious vulnerabilities
- IDS alerts and packet analysis together improve incident investigation accuracy

Visualization of Detected Attacks

An optional requirement of this project is the visualization of detected attacks using dashboards or graphs. This helps improve the interpretation of security events by presenting IDS alerts in a clear and structured format.

In this project, visualization was achieved through Wireshark traffic analysis and Suricata log review, which displayed network activity during attacks such as port scanning and SQL injection.

For enhanced monitoring, tool like EveBox was used to generate dashboards showing attack frequency, traffic patterns, and source IP activity, improving overall threat analysis and response.

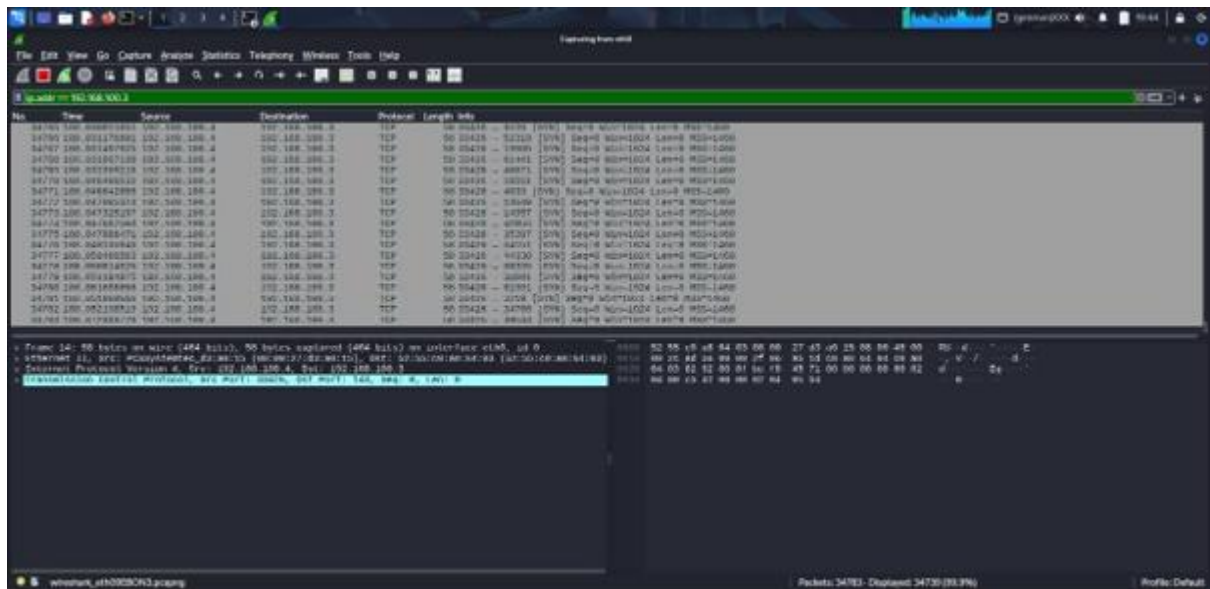
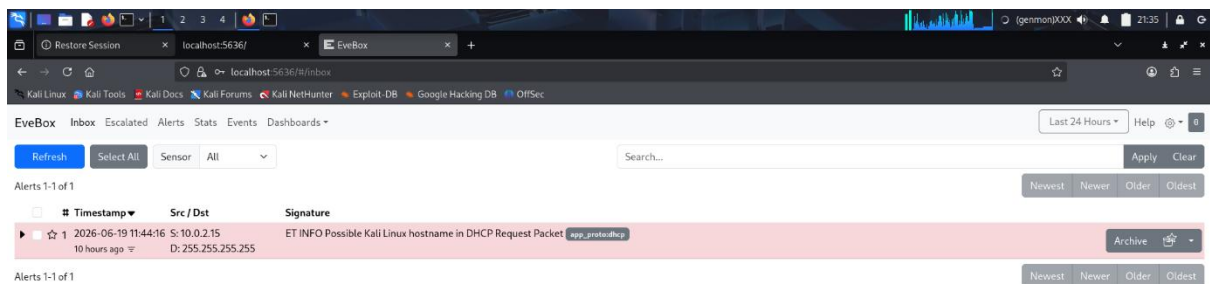


Figure 10: Wireshark capture of Suricata alerts



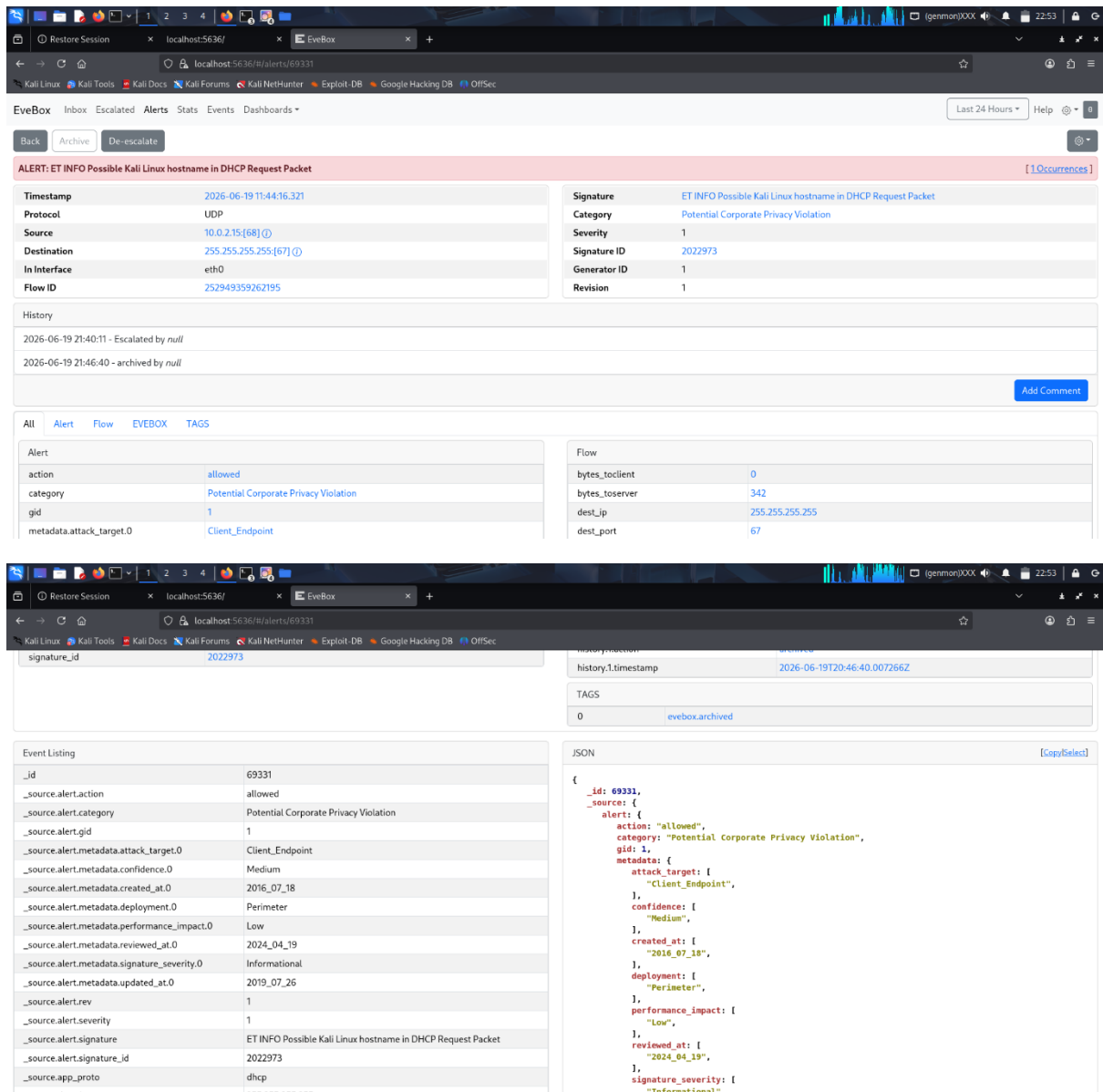


Figure 11: EveBox dashboard

Conclusion

This project successfully demonstrated the deployment and operation of an Intrusion Detection System using Suricata. Through simulated attacks, the IDS detected and generated alerts for malicious activities in real time. Packet analysis using Wireshark validated these detections by revealing underlying network behavior. The integration of IDS alerts, packet inspection, and response mechanisms highlights the importance of layered network security monitoring in modern cybersecurity environments.